

THATO BARUTI

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Scientific ML & AI Engineering · CFD / Simulation · Surrogate Modelling · Python / PyTorch · Validation-First Product Builder

Work authorisation note: Current UK Student visa; eligible to self-fund the UK Graduate Route after MSc completion, so no employer sponsorship is required for the initial post-study period.

PROFILE

AI and simulation engineer completing an MSc in Computational Fluid Dynamics at Cranfield University, with a BEng in Aerospace Engineering from the University of Sheffield. I build validated technical systems across scientific machine learning, CFD, optimisation, image analysis, and engineering decision support. My strongest pattern is taking an ambiguous technical problem from first principles to a usable artefact: data pipeline, model, baselines, tests, uncertainty, visual evidence, API or interface, and a clear explanation of where the system can and cannot be trusted.

WHAT I BRING

- Scientific ML and model reliability:** PyTorch/PyTorch Geometric models, graph neural networks, PINN-style physics losses, uncertainty calibration, external stress tests, failure-regime audits, and strict baseline comparisons.
- Simulation and engineering judgement:** OpenFOAM, ANSYS Fluent, UCNS3D, mesh generation, RANS validation, DNS post-processing, aerodynamic design, and numerical-methods reasoning.
- Product-minded AI delivery:** FastAPI and Gradio-style interfaces, JSON/CSV reports, overlays, confidence warnings, GitHub documentation, Docker/CI exposure, and stakeholder communication.
- Leadership under pressure:** led teams of 4 to 12 people in UAS, rocketry, and high-altitude aircraft projects, with quantified improvements in stability, endurance, drag, delivery speed, and manufactured prototype outcomes.

EDUCATION

MSc Computational Fluid Dynamics Oct 2025–Aug 2026 (Expected, on track for Distinction)
Cranfield University, UK
Data Analysis & Post-Processing (87%); Introduction to Fluid Mechanics & Heat Transfer (87%); Turbulence Modelling (72%); Numerical Methods & HPC; Grid Generation; Numerical Modelling.

BEng Aerospace Engineering Jun 2025
University of Sheffield, UK
Aerodynamic Design; Aircraft Dynamics & Control; Aerospace Design; Project Management.

Awards: Botswana Government Top Achievers Scholarship; University of Sheffield International Undergraduate Scholarship.

CORE TECHNICAL SKILLS

AI / ML	Python, PyTorch, PyTorch Geometric, scikit-learn, NumPy, model validation, graph neural networks, density models, uncertainty, baselines, failure analysis
Scientific SW	FastAPI, Gradio, Git/GitHub, Linux/HPC workflows, shell scripting exposure, CSV/JSON reporting, Docker/CI exposure, reproducible technical documentation
Simulation	OpenFOAM, ANSYS Fluent, UCNS3D, ParaView, Tecplot, Fidelity Pointwise, FluidX3D, RANS, DNS post-processing, turbulence statistics, mesh quality checks
Engineering	Aerodynamics, CFD validation, wing design, optimisation, UAS performance, force/moment analysis, experimental correlation, technical reporting
Developing	LLM and agentic AI workflows, cloud/MLOps, broader biomedical datasets, SQL/data engineering, and production CI/CD depth

SELECTED TECHNICAL EVIDENCE

ClusterCount-AI4S: Biomedical Cell-Counting Validation Framework 2026
Independent Study

- Built a validation-first biomedical image-analysis framework for automated cell counting under clustering, blur, and domain shift, framing cell counting as a measurement-validity problem rather than only a segmentation task.
- Implemented complementary model paths: an interpretable ClusterCountModel regressor for controlled operating-range benchmarking and a calibrated patchwise density pipeline for external microscopy adaptation.
- Evaluated on 300 held-out synthetic fluorescence-like images, achieving 4.67% MAPE versus 31.71% for a watershed baseline; error tightened to 4.45% inside a moderate blur and clustering envelope.
- Stress-tested on a 37-image external slice derived from BioStudies S-BSST265, reducing external MAPE to 34.3% after scale-normalised patchwise inference and lightweight calibration, while explicitly reporting remaining high-count and

cross-regime failures.

- Produced overlays, warnings, model-baseline disagreement, operating-range limits, tests, Docker/CI-style packaging, and a report grounded in Cellpose, StarDist, CellProfiler, BBBC resources, and uncertainty-under-shift literature.

AeroGenic: Scientific ML Platform for Aerodynamic Design

2025–Ongoing

MSc Research, Cranfield University

- Built graph-based, polar-structured, and hybrid ML models in PyTorch/PyTorch Geometric for aerodynamic coefficient prediction across wing design sweeps, combining geometry features, conditioning vectors, message passing, physics-aware losses, and target-wise model selection.
- Implemented the evidence spine: geometry-disjoint case-level splits, metadata-only baselines, shared metrics, parity plots, slice diagnostics, split-conformal uncertainty calibration, validation-case selection, model audits, and all-model reporting figures.
- Delivered strict held-out evidence: MGN/MGNM reached 3.18% MAPE for C_L , 2.60% for C_D , and 3.83% for C_{My} ; polar model improved C_D to 2.49%; target-wise champion ensemble achieved 2.87% deployment-weighted held-out MAPE.
- Built deployment-facing FastAPI and Gradio-style interfaces that convert geometry and flight requirements into traceable wing-design outputs, confidence scores, warnings, exported geometry, and JSON reports.

OpenFOAM RANS Validation and Residual Correction

2025–Ongoing

Cranfield University HPC Infrastructure

- Commissioned an automated RANS validation pipeline and fixed structural failures across mesh generation, surface feature extraction, solver stability, force-object compatibility, parallel decomposition, location-in-mesh selection, and remote-rank environment setup.
- Managed a 32-case validation campaign with 29 returned CFD cases and 26 strict-pass evidence cases, exposing the viscous drag gap inherited from lower-fidelity VLM-style labels.
- Built grouped geometry-held-out residual correction layers against higher-fidelity CFD, reducing corrected MAE to 0.0677 for C_L , 0.0104 for C_D , and 0.401 for C_{My} .

Requirement-Conditioned Pareto Optimisation

2025–Ongoing

Cranfield University

- Implemented an optimisation gate over 63,393 candidate designs, retained 15,031 feasible lift-band candidates, and extracted 1,012 non-dominated designs across drag, trim, lift overshoot, span, and training-manifold risk.
- Selected six representative designs for OpenFOAM re-simulation and used returned CFD results to separate promising lift-band survivors from surrogate-exploitation failures.

ADDITIONAL ENGINEERING AND ML EXPERIENCE

DNS Turbulent Channel Flow and Neural Prediction

2025

Cranfield University, Data Analysis & Post-Processing, 87%

- Processed high-fidelity DNS turbulent-channel data at $Re_\tau \approx 180$, computing RMS, skewness, kurtosis, spectra, and instantaneous-field relationships.
- Trained a feed-forward neural network to map velocity fields between spanwise planes and benchmarked outputs against held-out ground truth and Lee & Moser style reference data.

Godunov Compressible-Flow Solver Benchmarking

2025

Cranfield University

- Extended and benchmarked a Fortran finite-volume solver with MUSCL reconstruction and HLL/HLLC Riemann solvers for 1D and 2D Euler equations, validating on Mach 3 forward-facing step and double Mach reflection cases in UCNS3D.

NASA Trap Wing High-Lift Grid Generation

2025

Cranfield University

- Generated high-quality hybrid meshes in Fidelity Pointwise for the NASA Trapezoidal High-Lift Wing, targeting wall-resolved RANS with $y^+ \approx 1$, boundary-layer inflation, and wake refinement.

Local AI and Agentic Workflow Exploration

2026

Independent Learning

- Used LM Studio and local Gemma models to explore privacy-friendly LLM workflows for scientific reporting, validation summaries, technical assistant behaviour, and tool-oriented AI patterns.

LEADERSHIP AND TEAM DELIVERY

AeroMAP UAS Build

Oct 2023–May 2024

Performance Lead, University of Sheffield

- Led a four-person performance team using simulation and test evidence to improve stability by 25% and endurance by 15%, translating design trade-offs for a multidisciplinary group.

SunStratos High-Altitude Surveillance Glider

Feb 2023–Jun 2024

Structures Lead, University of Sheffield

- Led a 12-member team from concept through manufactured prototype, improving delivery speed by 33% through task

ownership, scheduling, and clear communication.

UKSEDS National Rocketry Competition
Design Team Lead, University of Sheffield

Oct 2022–May 2023

- Led aerodynamic analysis and design iteration, reduced drag coefficient by 22%, and ranked Top 10 nationally among 30+ UK university teams.

ROLE FIT

Best fit now	Scientific ML engineer, applied ML engineer, CFD/simulation engineer, research engineer, surrogate-modelling engineer, AI-for-engineering platform builder, technical analyst
Strongest proof	ClusterCount-AI4S, AeroGenic, OpenFOAM validation, MGN/polar/hybrid models, Pareto optimisation, FastAPI/Gradio-style reporting, UAS/rocketry leadership
Working style	Evidence-driven, fast learner, strict about limitations, comfortable with ambiguous technical problems, able to explain complex systems to non-specialists
Next growth	Production cloud/MLOps, larger direct software-engineering codebases, LLM/agent systems, broader biomedical AI, and industrial validation experience